Practice for Quiz 1 IEORE4004, Fall 2025
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Problem 1

Your clothing company owns one warehouse, three shops—call them A, B, and C—and sells three products: trousers, skirts, and pullovers. Your current stock at the warehouse is:

Trousers:	200 units;
Skirts:	80 units;
Pullovers:	250 units.

Each unit of the three products has a certain degree of flammability:

Trousers:	level 3;
Skirts:	level 2;
Pullovers:	level 5.

By law, the sum of the degrees of flammability of products in your shops cannot exceed the following:

Shop A: 300; Shop B: 400; Shop C: 500.

The profit you make by selling a unit of these products is:

Trousers: 10\$; Skirts: 5\$; Pullovers: 7\$.

1. Assuming that all products placed in your shops will be sold and there is no cost for shipping products from the warehouse to the shops, formulate an LP that decides the quantity of each product that will be shipped from the warehouse to each of your shops so as to satisfy the constraints above and to maximize your profit (ignore the integrality constraints on the variables).

2. Now assume the following unitary shipping costs:

$$\text{Trousers: } \begin{cases} \text{to shop A: } 4 \\ \text{to shop B: } 3 \\ \text{to shop C: } 3 \end{cases} \qquad \text{Skirts: } \begin{cases} \text{to shop A: } 1 \\ \text{to shop B: } 2 \\ \text{to shop C: } 3 \end{cases}$$

$$\text{Pullovers: } \begin{cases} \text{to shop A: } 2 \\ \text{to shop B: } 1 \\ \text{to shop C: } 2 \end{cases}$$

   How does your LP change?

3. Now assume that the constraint on the total degree of flammability is replaced by a constraint on the *average* degree of flammability of products that cannot be exceeded in the shops:

   Shop A: 3.8; Shop B: 4.2; Shop C: 4.

   How does your LP change?

Problem 2

Problem 4, part a) from Assignment 2.

Problem 3

Problem 5 from Assignment 2.

Problem 4

Problem 1 from Assignment 3.

Problem 5

Problem 3 from Assignment 3.

Bonus question

Problem 4, part c) from Assignment 3.

We report here the solution to Problem 1. The others can be found in the solutions to the corresponding assignments.

1. If we denote by x_{tA} the number of trousers shipped from the warehouse to shop A, and similarly $x_{tB}, x_{tC}, x_{sA}, x_{sB}, x_{sC}, x_{pA}, x_{pB}, x_{pC}$, we have:

$$\begin{aligned}
\max \quad & 10(x_{tA} + x_{tB} + x_{tC}) + 5(x_{sA} + x_{sB} + x_{sC}) + 7(x_{pA} + x_{pB} + x_{pC}) \\
\text{s.t.} \quad & x_{tA} + x_{tB} + x_{tC} \leq 200 \\
& x_{sA} + x_{sB} + x_{sC} \leq 80 \\
& x_{pA} + x_{pB} + x_{pC} \leq 250 \\
& 3x_{tA} + 2x_{sA} + 5x_{pA} \leq 300 \\
& 3x_{tB} + 2x_{sB} + 5x_{pB} \leq 400 \\
& 3x_{tC} + 2x_{sC} + 5x_{pC} \leq 500 \\
& x \geq 0
\end{aligned}$$

2. Add to the objective function:

$$-4x_{tA} - 3x_{tB} - 3x_{tC} - x_{sA} - 2x_{sB} - 3x_{sC} - 2x_{pA} - x_{pB} - 2x_{pC}.$$

3. Replace constraints 4 to 6 in the LP with

$$\begin{aligned}
3x_{tA} + 2x_{sA} + 5x_{pA} &\leq 3.8(x_{tA} + x_{sA} + x_{pA}) \\
3x_{tB} + 2x_{sB} + 5x_{pB} &\leq 4.2(x_{tB} + x_{sB} + x_{pB}) \\
3x_{tC} + 2x_{sC} + 5x_{pC} &\leq 4(x_{tC} + x_{sC} + x_{pC})
\end{aligned}$$